

Dylan Smith

CS 418 Group Project Proposal

My proposed project is titled “How much does NVIDIA affect the rest of the tech sector?”

I trade stocks semi-frequently and have noticed that many of the tech companies I invest in have their stock prices moved based on how NVIDIA's stock is moving. My github handle is dsmith513. I do not currently have any proposed teammates in mind but would be open to working with anyone with a financially based project proposal.

I invest in a lot of smaller market tech companies and have noticed that a lot of their stock movement reflects NVIDIA's stock movement. Over the past few years, especially with the rise of the AI industry, it has seemed as if NVIDIA has dictated the direction of the tech sector recently. With so many new data centers being constructed that run exclusively NVIDIA GPUs for AI training, it seems inevitable that NVIDIA and the AI industry will dominate the tech sector for at least the near future. I want to investigate just how closely NVIDIA's stock price reflects the stock prices of other tech companies, and I hypothesize that they will be closely related.

The dataset that I plan to use is located on Hugging Face and can be accessed for free by anyone. This is the dataset repository: [paperswithbacktest/Stocks-Daily-Price](#). It contains the symbol, date, opening price, highest price, lowest price, closing price, and adjusted closing price of over 7,000 US stocks going back to as early as 1962 for some stocks. The dataset is quite large, containing 24.8 million rows and is designed for training AI models, but the data can be accessed easily via SQL queries to make collecting specific portions of data easy. Additionally, the dataset is updated daily to contain that day's stock information regarding the stocks in the dataset.

I plan to approach the problem by comparing the growth of NVIDIA's share price with the share price of other tech companies over the last 3 years. I will then cross-examine this growth with the growth of other major industries in the same time-frame. The end result will likely be a visualization tool that will allow users to compare NVIDIA's growth with specific stocks and overall industries. The data can be analyzed using regression modeling or heatmaps to visualize these comparisons. By the progress report, I hope to have a clear direction for the project as well as a composition of all the relevant data I will need to complete the project.